

120 Days of Quant Finance

Week 7: Time Series Analysis Basics

Day 44: Stationarity and Time Series Properties

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1. Definition of Stationarity

A time series $\{X_t\}$ is stationary if its statistical properties do not change over time.

2. Strict Stationarity

A process is strictly stationary if for any time shift h :

$$(X_{t_1}, X_{t_2}, \dots, X_{t_k}) \stackrel{d}{=} (X_{t_1+h}, X_{t_2+h}, \dots, X_{t_k+h})$$

for all $t_1, \dots, t_k$.

3. Weak (Covariance) Stationarity

A process is weakly stationary if:

- $E[X_t] = \mu$ (constant mean)
- $Var(X_t) = \sigma^2$ (constant variance)
- $Cov(X_t, X_{t+k}) = \gamma(k)$ (depends only on lag)

4. Examples

White Noise

$$X_t = \epsilon_t$$

where $E[\epsilon_t] = 0$ and $Var(\epsilon_t) = \sigma^2$.

White noise is stationary.

Random Walk

$$X_t = X_{t-1} + \epsilon_t$$

The variance increases over time, making the process non-stationary.

5. Autocovariance and Autocorrelation

Autocovariance function:

$$\gamma(k) = Cov(X_t, X_{t+k})$$

Autocorrelation function:

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)}$$

Conclusion

Stationarity ensures that a time series has stable statistical properties over time. It is a fundamental requirement for most forecasting and econometric models.